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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	Hoo Lee Sheet
Project / Solution Name	Laluan
Name (Leader)	Tan Bing Jie
Email & Phone (Leader)	tanbingjie99@gmail.com 012-8002983
University Affiliation(s)	Xiamen University Malaysia (XMUM)
Live Prototype URL	https://hoo-lee-sheet-sdc-2026.replit.app (Replit, main). Backup: https://laluan.vercel.app
Code Repository URL (public / access granted to judges)	https://github.com/Jie-coder/hoo-lee-sheet-sdc2026 (public)

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Laluan is a Next.js (React) web app. It runs on Replit as the main deployment, with Vercel as the backup. The user uploads a CV or picks a sample CV. A language model (Claude Haiku 4.5 on Replit, Gemini on Vercel) parses it into skills and an occupation. gemini-embedding-001 matches the skills to ESCO skills. The app then reads exposure scores from ILO WP140 occupation averages, salary data from DOSM/OpenDOSM, scenario parameters from WEF, and courses from a curated catalogue, and shows the results with their sources. Data is stored in Supabase (Postgres with pgvector). AI-trend signals come from an arXiv fetch script (run manually) that stores dated snapshots.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Next.js 16.4.2; React / React DOM 19.3.0 (Node >=24)	Web app framework and UI library	MIT
Tailwind CSS ^4; shadcn/ui (shadcn ^4.22.0); @base-ui/react ^1.9.0; lucide-react ^1.58.0; framer-motion ^14.4.0	Styling, UI components, icons and animation	MIT (lucide-react: ISC)
Vercel AI SDK: ai ^7.0.143, @ai-sdk/anthropic ^4.0.85, @ai-sdk/google ^4.0.99, @ai-sdk/openai-compatible ^3.0.71	Calls to the language and embedding models (Claude Haiku, Gemini)	Apache-2.0
@supabase/supabase-js ^2.118.0; pgvector (Postgres extension on Supabase)	Database client for Supabase (Postgres); vector search for skill matching	MIT (Supabase JS); PostgreSQL License (pgvector)
zod ^4.6.7; unpdf ^1.9.1	Schema validation (Zod); PDF text extraction for uploaded CVs (unpdf)	MIT

Name & Version	Purpose in Project	License
Dev: TypeScript ^5, Vitest ^5.0.5, ESLint ^9	Development only: language, tests (final suite 510 tests), linting	Apache-2.0 (TypeScript); MIT (Vitest, ESLint)
starter-v0 (private repo Jie-coder/sdc-starter, tag starter-v0, commit 57171fe)	Pre-existing team starter code, imported in the initial repo setup; use approved by organiser Jason on 17 Oct 2026	Team's own pre-existing code (not third-party); repository is private

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Google Gemini gemini-3.8-flash (Vercel backup and fallback)	Chat model on Vercel and fallback on Replit via getModel(): CV parsing; closest-occupation pick only when exact-name and alias matches fail. Explanations, scores and pathways come from code (no model call)	Free (team leader's own API key)
Google gemini-embedding-001 (skill matching)	Skill and role embeddings	Free API key
Anthropic Claude Haiku (claude-haiku-4-5) via Replit AI Integrations	Chat model on Replit via getModel(): CV parsing; closest-occupation pick only when exact-name and alias matches fail (PR #48)	Sponsor-provided (billed to organiser-provided Replit Core credits)
arXiv API	AI-trend signal: fetch script (scripts/fetch-arxiv.ts) run manually, dated snapshots	Free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Deployment, main hosting (compulsory), from the team leader's own account	Replit Core (sponsor), Autoscale, Asia
Vercel	Backup / preview deployment	iad1 (Washington, D.C., USA), default
Supabase ("Hoo Lee Sheet's Project")	Database + pgvector	Free tier, ap-southeast-1 (Singapore)
GitHub	Repo (github.com/Jie-coder/hoo-lee-sheet-sdc2026); pull-request review (no GitHub Actions or CI)	Free

Service & Provider	Purpose in Project	Plan / Region
RunPod	Not used (committee confirmed 21 Oct that RunPod is only required for GPU work; Laluan uses hosted model APIs with no self-hosted GPU)	n/a

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Anthropic Claude Haiku 4.5 (claude-haiku-4-5)	Anthropic, via Replit AI Integrations (organiser Replit Core credits); main model for CV parsing and occupation matching on Replit	Pre-trained, called via API (no training or fine-tuning by the team)	https://www.anthropic.com/legal/commercial-terms
Google Gemini gemini-3.8-flash	Google, team's own API key; fallback model on the Vercel deployment	Pre-trained, called via API (no training or fine-tuning by the team)	https://ai.google.dev/gemini-api/terms
Google gemini-embedding-001	Google; embeddings for skill matching	Pre-trained, called via API (no training or fine-tuning by the team)	https://ai.google.dev/gemini-api/terms
LLMs used in development only (by Grok): Claude models; DeepSeek	Claude models via Claude Code and Cursor cloud agents; DeepSeek via OpenCode	Pre-trained, used through the tools in 4.3; not part of the product	n/a (development tools only; see 4.3)

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
OpenDOSM open data (wages_occupation)	OpenDOSM; https://open.dosm.gov.my/data-catalogue/wages_occupation; fetched by scripts/fetch-dosm-demand.ts (PR #30)	Public	CC BY 4.0	N
JobStreet Malaysia Career Advice public salary pages	https://my.jobstreet.com/career-advice, curated by hand into data/salary_by_role.csv (no scraping, no API); salary card only, not scoring; 32 of 40 roles (PR #50)	Public	Used. The team checked salary figures by hand on the JobStreet platform as a reference; no scraping, no API. Terms: personal, non-commercial use; https://my.jobstreet.com/terms.	N
DOSM salary survey major-group averages	Fallback salary for the other 8 roles (PR #50, commit 47fbbcf)	Public	Government Open Data Terms of Use 1.0 (DOSM open data portal), source: https://www.dosm.gov.my/portal-main/article/term-of-use; attribution: "Source: Department of Statistics Malaysia".	N
Three synthetic sample CVs	Created by the team with AI assistance (PR #3)	Synthetic	Apache-2.0	N (synthetic)
ILO Working Paper 140 occupation exposure scores	https://www.ilo.org/sites/default/files/2025-05/WP140_web.pdf	Public	CC BY 4.0	N
ESCO v1.2 occupations and skills	https://esco.ec.europa.eu/en/about-esco/escopedia/escopedia/frequently-asked-questions	Public	Reuse with acknowledgement under Commission Decision 2011/833/EU	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Anthropic Economic Index (release 2026-06-26)	https://huggingface.co/datasets/Anthropic/EconomicIndex	Public	Dataset card says "Data released under CC-BY, code released under MIT License" (no CC-BY version stated)	N
WEF Four Futures for Jobs (2026)	https://reports.weforum.org/docs/WEF_Four_Futures_for_Jobs_in_the_New_Economy_AI_and_Talent_in_2030_2025.pdf	Public	Report prints "© 2026 World Economic Forum. All rights reserved"; WEF's general CC BY-NC-ND 4.0 terms apply only "unless otherwise indicated" (https://www.weforum.org/about/licence-terms-on-the-use-of-forum-publications-and-materials/); used as cited parameters only.	N
arXiv metadata	arXiv API (https://arxiv.org/help/api/index)	Public	CC0	N
Course catalogue links (Coursera, Alison, e-LATiH, Microsoft Learn, Cisco Networking Academy, OpenLearn, HRD Corp, MyFutureJobs, smaller providers)	Public provider pages (links only)	Public	Under each provider's terms: Coursera, Alison, e-LATiH, Microsoft Learn; Cisco Networking Academy (Terms and Conditions, https://www.netacad.com/resources/legal/terms-conditions); OpenLearn (CC BY-NC-SA 4.0 unless otherwise stated, https://about.open.ac.uk/strategy-and-policies/policies-and-statements/copyright-openlearn); HRD Corp website and MyFutureJobs: Licence not stated; used for citation/reference only (source: https://hrdcorp.gov.my/; https://myfuturejobs.gov.my/); smaller single-course providers: each provider's own site terms apply, not checked individually.	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Claude Code	Writing code (from T06 onwards), operated via Grok	Major
Cursor cloud agents	Writing code (T03, T05, scoring engine, T04a results page; PRs #2 onward), operated via Grok	Moderate
Grok	(a) technical plan; initial repo setup commits (starter import, Apache 2.0 licence, Replit config, RunPod option, README); PR #5 (data/README.md); code review and merge of all PRs; (b) docs and rules/compliance checks; (c) dispatching coding tasks and opening PRs; (d) design: design system, Laluan logo, results page reference, wireframes, 3 synthetic CVs (PR #3, #4; synthetic CVs are AI-assisted demo data); pitch deck rendering via its own Python HTML-to-deck script; hand-drawn line SVG illustrations	Major
OpenCode (DeepSeek model)	Light edits from 23 Oct while Claude Code was rate-limited: commits 9729220 (i18n strings) and 0f31c5b (handleDownload + first draft of pathway print receipt) on branch jarvis/caldera-redesign; PRs #56, #57, #58 partly (rest hand-written). Print CSS and demo PDF deletion were hand-written.	Minor

Tool & Version	Used For	Extent of Use (minor / moderate / major)
Google Stitch (Google Labs): AI UI design tool that turns text prompts and images into UI designs and front-end code, built on Gemini. https://stitch.withgoogle.com/; https://developers.googleblog.com/stitch-a-new-way-to-design-uis/	UI/visual inspiration for the website design (no generated code used in the product)	Minor

Other design resources: Google Fonts Anton, DM Sans, DM Mono (pitch deck typography, open-source licences); Caldera used as visual-style inspiration only, no Caldera assets used; demo visuals are screenshots of Laluan itself.

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Grok's code changes all went through GitHub pull requests. A human read and reviewed every PR before it was merged: Grok gave a first-pass code review, and Bing Jie reviewed and approved every merge. There is no CI pipeline; the final Vitest suite passed 510 tests. The team tested the app by hand on the live Replit and Vercel deployments with the sample CVs. Model output in the product is limited to CV parsing and occupation matching from a fixed list; scores, salaries and course links come from cited data files, not from a model.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit	Replit Core (Handbook 9.1.1)	Main hosting of the live prototype; Claude Haiku via Replit AI Integrations billed to these credits
RunPod	US$25 credit (Handbook 9.1.2)	Not used

Proof of Replit use (Handbook 10.6). Replit plan: Replit Core (claimed). Live deployment https://hoo-lee-sheet-sdc-2026.replit.app, on Replit project hoo-lee-sheet-sdc2026 under account TANBINGJIe99, linked to the GitHub repo Jie-coder/hoo-lee-sheet-sdc2026. Screenshots (also in the team's Final Submission folder):

Figure 1. Replit Publishing page showing the deployment as Live: https://drive.google.com/file/d/1RTCqEs8HdV3HVjSvUJUM03fDsPmRWwNZ/view

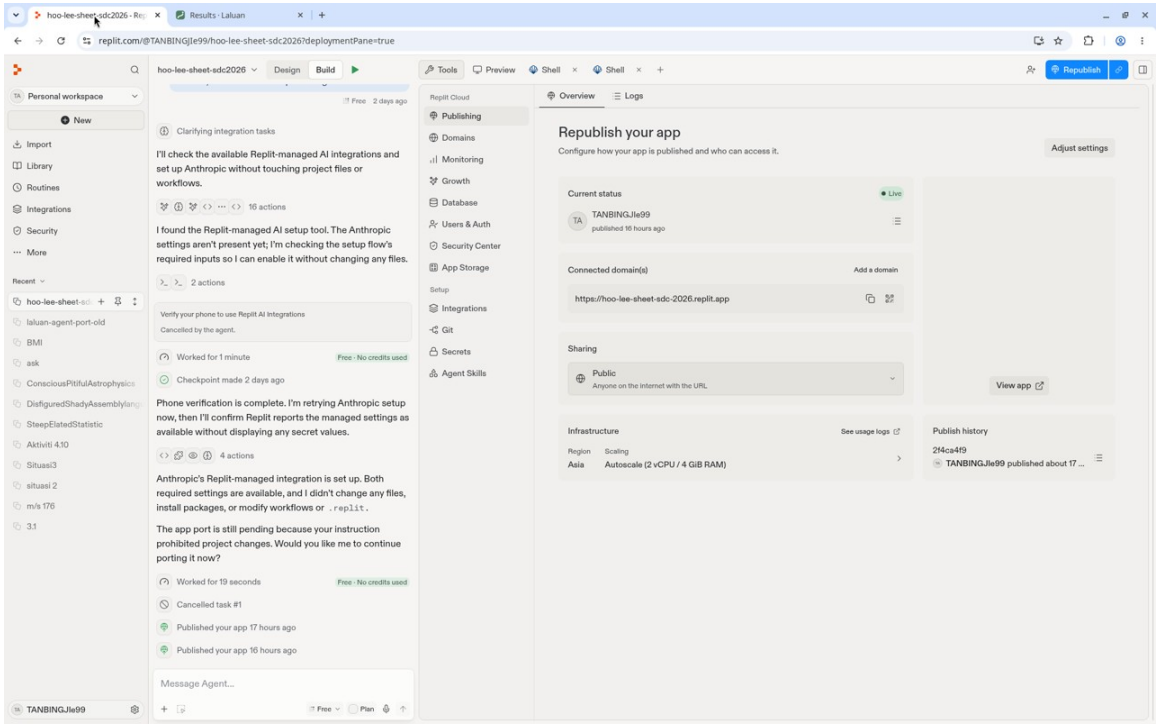


Figure 2. Replit usage page, billing period 22 Oct to 21 Nov, $2.32 used: https://drive.google.com/file/d/1R-yi-V7Nh6n6SYa0RaRWYfqlO5yXCK-K/view

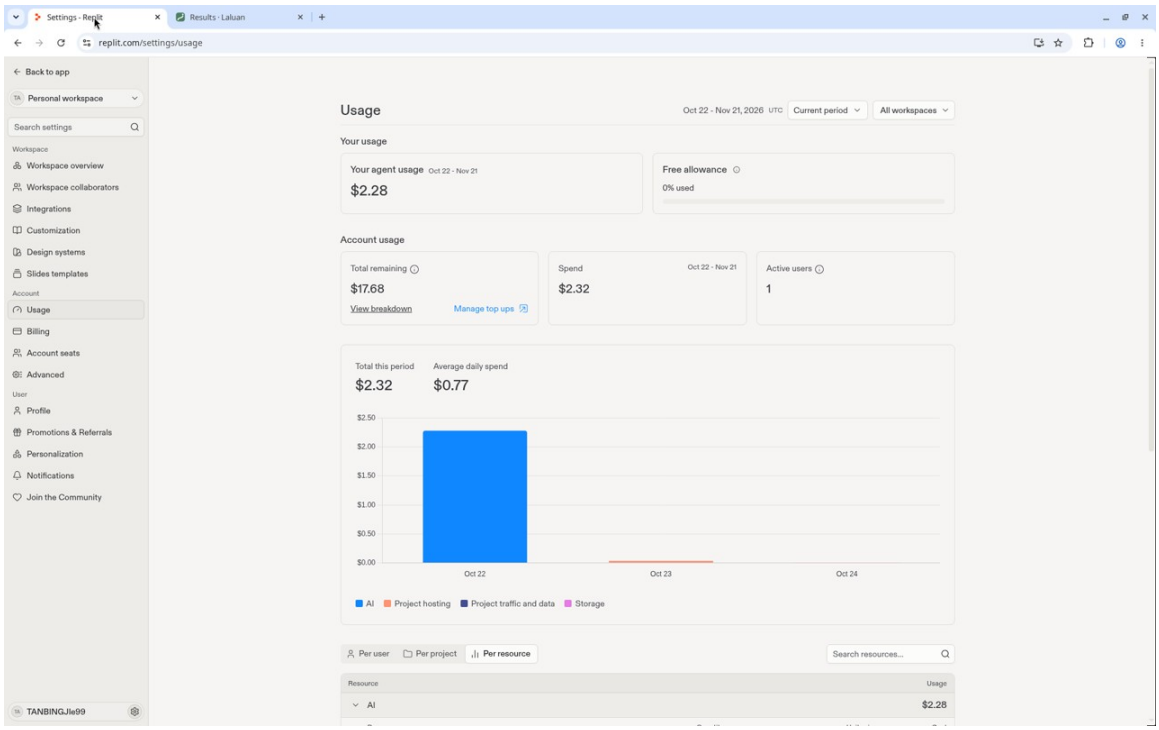


Figure 3. Git history of the Replit project, linked to the GitHub repo: https://drive.google.com/file/d/17TcbC3jp5SsijE4GOYXuq50n3MO0WAA9/view

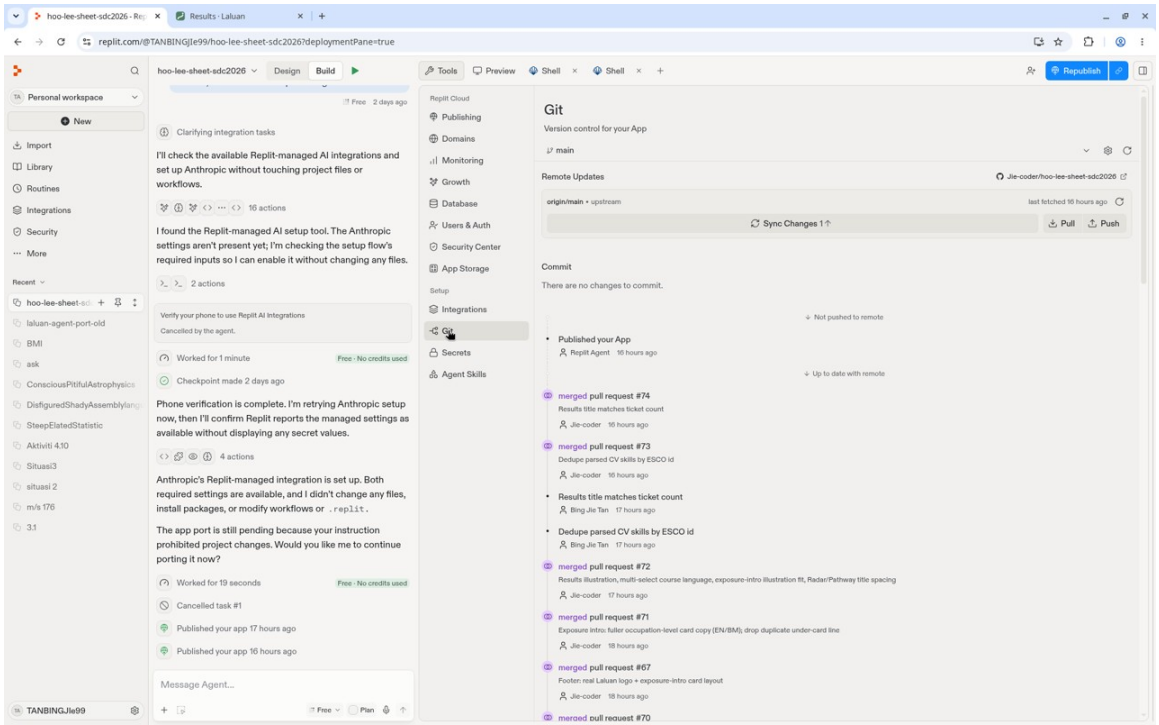


Figure 4. The live Laluan app at the Replit URL: https://drive.google.com/file/d/1pZw8tuTBIRno9HuWp6M90_3zir5qK3Tn/view

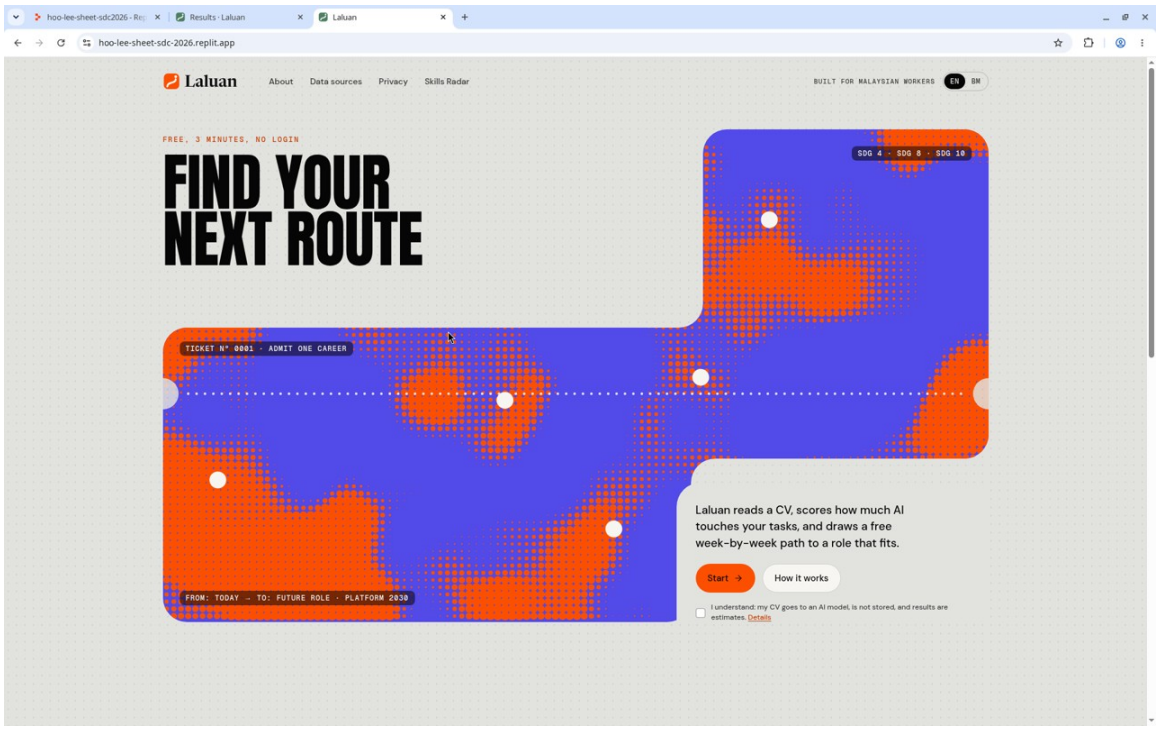
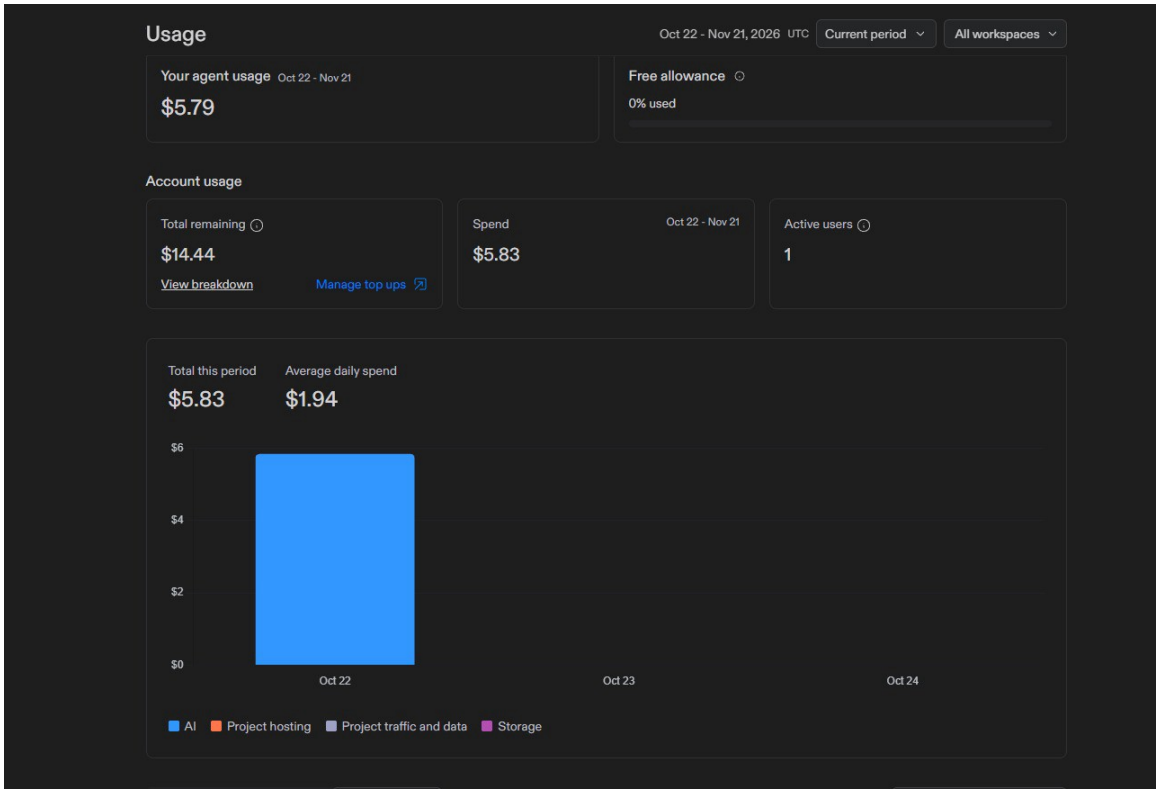


Figure 5. Usage on teammate Henry's Replit account, $5.83 spent: https://drive.google.com/file/d/1Knn3OhWhW6HbCYLD05wNxBc6jQh0Rgd2/view



Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit (Replit Core credits from the organiser) is the main host and pays for Claude Haiku 4.5 through Replit AI Integrations, so it is used as the primary platform. It is replaceable: the same code runs on Vercel as a backup (https://laluan.vercel.app), with Google Gemini on the team's own key as the fallback model, and the pre-parsed sample CVs keep the demo working without any live model call.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Laluan has no login and no accounts. An uploaded CV is sent once to the language model (Claude Haiku 4.5 on Replit, or Gemini on the Vercel backup) for parsing into skills and is not stored by Laluan. Users see and can edit every parsed skill before results are shown. The demo uses three synthetic sample CVs that are pre-parsed, so no real personal data is needed to judge the prototype. API keys are kept in platform secrets, never in the repository or this document. Exposure scores are computed from published ILO data, not by a model, and the app states that its outputs are estimates to support career decisions, not hiring decisions.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

All scores and forecasts are estimates and are labelled as such. AI exposure is not job loss. The WEF scenarios are an illustrative parameter table (3 of 4 futures used). Only 40 ESCO roles are covered, and course and salary data are checked by hand on a stated date, so they can go out of date. The model can mis-read a CV; users can edit every parsed skill. Some programmes have eligibility rules we cannot check, so the app shows "please check eligibility yourself". Laluan supports career decisions and never makes hiring decisions. If the model API, network or a free tier fails, a DEMO_MODE cache of the three synthetic personas keeps the demo running, with Vercel as backup host.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team’s knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



Signature:

Date: 24/10/2026